

Wall-E

Upgrade project by Richard Nicholson

Start: 16-5-26

Summary: -

This project is based on the original work done by Chillbasket. Link below.

Base code was taken from the Wall-E ESP32 remix. Link below

A few upgraded Wall-E prints were located and used. The complete set is in this download.

- Upgrades: -
 - Web Server removed – has major memory leak issue
 - Added BLE Gamepad controller – Xbox control 2026
 - Added working Top buttons
 - Added Bay Door Servo. Opens Closes on servo control
 - Controllable Bay LED light
 - Electronics Platform – frees up Bay so can have a boot and plant etc
 - RGB mood LED
 - 1.44 Oled feedback display
- 10 DOF:-
 - Arms (2)
 - Neck (2)
 - Head Rotation (1)
 - Eyes (2)
 - Eye Borrows (2)
 - Bay Door (1)
- **Size:** H = 338 W = 291 D = 186
- **Wall-E Assembly Instructions and Videos by Chillbasket**
<https://wired.chillibasket.com/3d-printed-wall-e/>

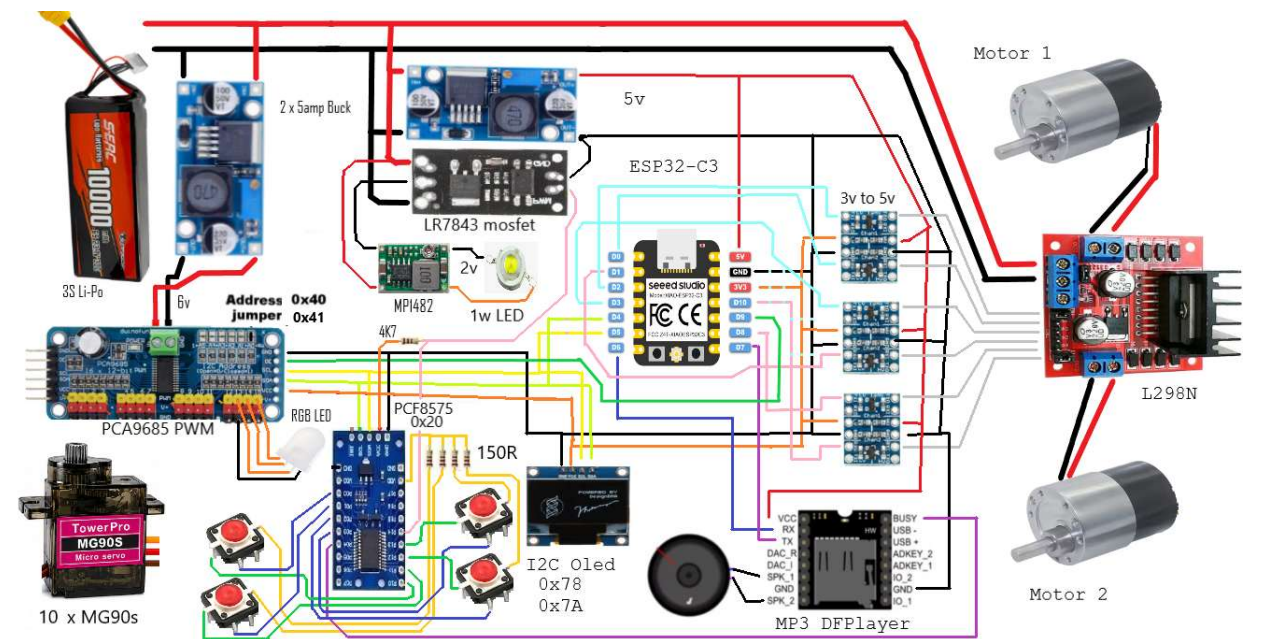
Hardware: -

- 1 x ESP32-C3
- 1 x I2C PCA9685 16-Channel 12-bit PWM Servo Motor Driver Module
- 1 x I2C PCF 8575 I/O Multiplexer
- 1 x Audio Module - Mini MP3 Player Audio Module (DFPlayer)
- 1 x mini-Speaker 8 or 4 ohm 3watt
- 1 x 1.44 I2C OLED Display
- 1 x 10mm RGB LED – common cathode
- 4 x Tactile Pushbutton with LED
- 4 x 150R resistor
- 1 x 4K7 Resistor
- 3 x Logic Level Shifter 3v to 5v
- 1 x L298N Dual H Bridge DC Motor Driver
- 10 x MG90S metal gear servos
- 2 x 37mm motor with gearbox, offset shaft, 25kg, 150 rpm (Not cheap ones)
- 1 x LR7843 Mosfet module
- 1 x MP1482 min buck set to 2v
- 1 x 1watt or 3watt LED (torch head light LED)
- 2 x Buck 5amp Set to 5v and 6v
- 1 x 10amp auto reset fuse
- 1 x XT60 Male Connector
- 16 x Bearings 10 x 15 x 4
- Misc. M3 bolts, nuts, heat inserts, heat shrink

Servos: -

- 10 x MG90S metal gears (there the mini ones)
- Stall torque: 1.8kg/cm (4.8V); 2.2kg/cm (6.6V)
- Dimension: 22.8×12.2×28.5mm
- Note: these servos are very weak. The arms and lower neck could be replaced with MG995 Servos.

Circuit: -



Servo Channel List: -

- 0 Head Rotation
- 1 Neck Top
- 2 Neck Bottom
- 3 Eye Right
- 4 Eye Left
- 5 Arm Left
- 6 Arm Right
- 7 Eye Brow Right
- 8 Eye Brow Left
- 9 Door
- 10
- 11
- 12
- 13 RED RGB LED
- 14 GRN RGB LED
- 15 BLU RGB LED

Buttons Functions: -

- RED PB - Play Prev
- GRN PB - Play Next
- ORG PB - Autonomous Servos On/Off
- WHT PB - Door Open/Close

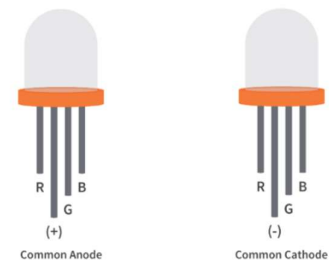


Figure 1 RGB LED. Use a Common Cathode LED

BLE Gamepad - Xbox controller - Button Map :-

Tracks turn left/right	Left Stick X (hoz)
Tracks forward/back	Left Stick Y (Vert)
Head rotation	Right Stick X
Head Up/Down	Right Stick Y
Neck Bottom	Button A + Dpad Up/Dwn
Lhs Eye	Button X + Dpad Up/Dwn
Lhs Eyebrow	Button X + Dpad Left/Right
Rhs Eye	Button B + Dpad Up/Dwn
Rhs Eyebrow	Button B + Dpad Left/Right
Door Open/Close	Button Y + Dpad Left Right
Left Arm	Left Trigger + Dpad Up/Dwn
Right Arm	Right Trigger + Dpad Up/Dwn
Play Sound	Left Bumper + Dpad Left/Right - can crash
Motor dead zone	Left Bumper + Dpad Up/Dwn - 0 to 250
LEDS move	Right Bumper + Dpad Left/Right - loop x 4
Steering Offset	Right Bumper + Dpad Up/Dwn - -100 to 100
Animation Number	ViewButton + Dpad Up/Dwn
Automonus Mode	MenuButton On/Off toggel
Dpad Up	adjust up
Dpad Down	adjust down
Dpad Left	adjust left
Dpad Right	adjust right
Left Stick Button	
Right Stick Button	
ShareButton	
XboxButton	

Notes

All moving joints need to be sanded and made to work as smoothly as possible.

Silicon from (Starlink Installations) is useful on these joints

Power up PSU and adjust Buck to 5v and 6v and 2v BEFORE connecting to main loom

Wire up loom and test before securing to 3d prints

Secure electronics to 3d prints, power up with no Motor or Servo loads, test buttons, leds, oled, MP3 player, xbox controller is working.

Connect servos and put calibration firmware in ESP32 and perform calibration (see chillibasket assembly link above)

Edit main wall-E firmware to add the calibration values from IDE serial monitor, transfer to ESP32 (Library's required to compile, see below)

Power up Wall-E

Turn on Game controller, should enrol automatically, watch Oled

Test and go for gold

Note: There's no Battery monitor in this code. Use a Li-Po battery monitor/buzzer on Li-Po monitor plug



Figure 5 RGB Front LED on PWM can do any colour (mood LED)

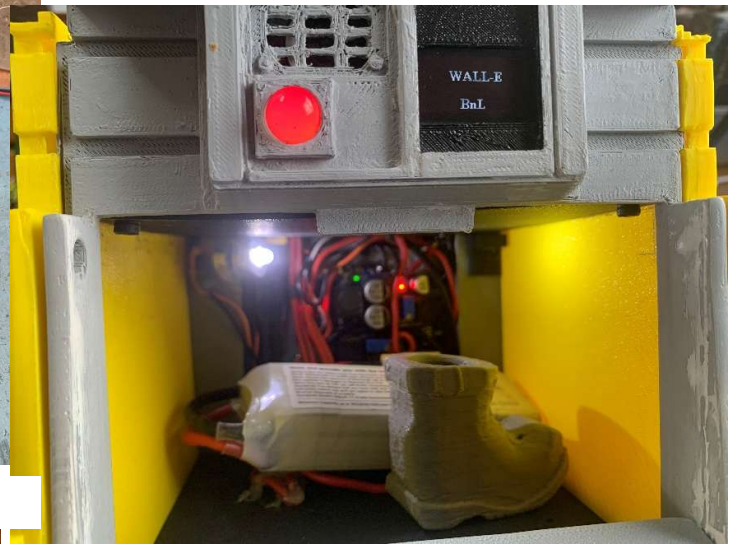


Figure 4 The White Bay LED is controllable



Figure 3 Electronics space



Figure 2 3 x Buck PSU, Mosfet, Auto Reset Fuse

Arduino IDE will require these library's

BLE-Game_Client by Tomaz bekas V0.12.1
ESP_Software_Serial by Dirk Kaar SNDVAPS 8.3.1
Adafruit SSD1306 by Adfruit v2.5.17
Adafruit PWM Servo Driver lib by Adfruit v 3.03
Adafruit PCF8574 by Adfruit v 1.1.2
DFRobotDFPlayerMini by DFRobot v1.0.6
Wire.h - I2C library driver, Could be built into IDE Library's

Note: at time of my build the BLE-Game_Client required a small edit in a txt file see link below to fix. It will be fixed in newer version releases, so this edit will not always be required.

Links: -

<https://wired.chillibasket.com/3d-printed-wall-e/> - Wall-E Chillbasket assembly instructions

<https://www.youtube.com/watch?v=QidMAAtZf88> – Wall-E Vid Example

<https://github.com/tbekas/BLE-Gamepad-Client/pull/122> - Game Pad – How to fix lib

<https://www.youtube.com/watch?v=mHWRYybAbvU> – Painting Wall-E for effect

Example Tracks Motors: -

<https://www.robotshop.com/products/dyna-engine-12v-600rpm-brushed-dc-gear-motor-w-37mm-diameter-gearbox?srsId=AfmBOooWpu5g35f-Ok1rgoiM7nlfjCGTsUTBmkXb6i3MAQAp3cv1vgnl>

Mr Prints Wall-E (Chillbasket): <https://www.printables.com/model/408363-wall-e/remixes>

Wall-E With ESP32 remix: <https://www.printables.com/model/852117-wall-e-with-esp32>